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(54) **HETEROGENEOUS INTEGRATION OF
RADIO FREQUENCY TRANSISTOR
CHIPLETS HAVING INTERCONNECTION
TUNING CIRCUITS**

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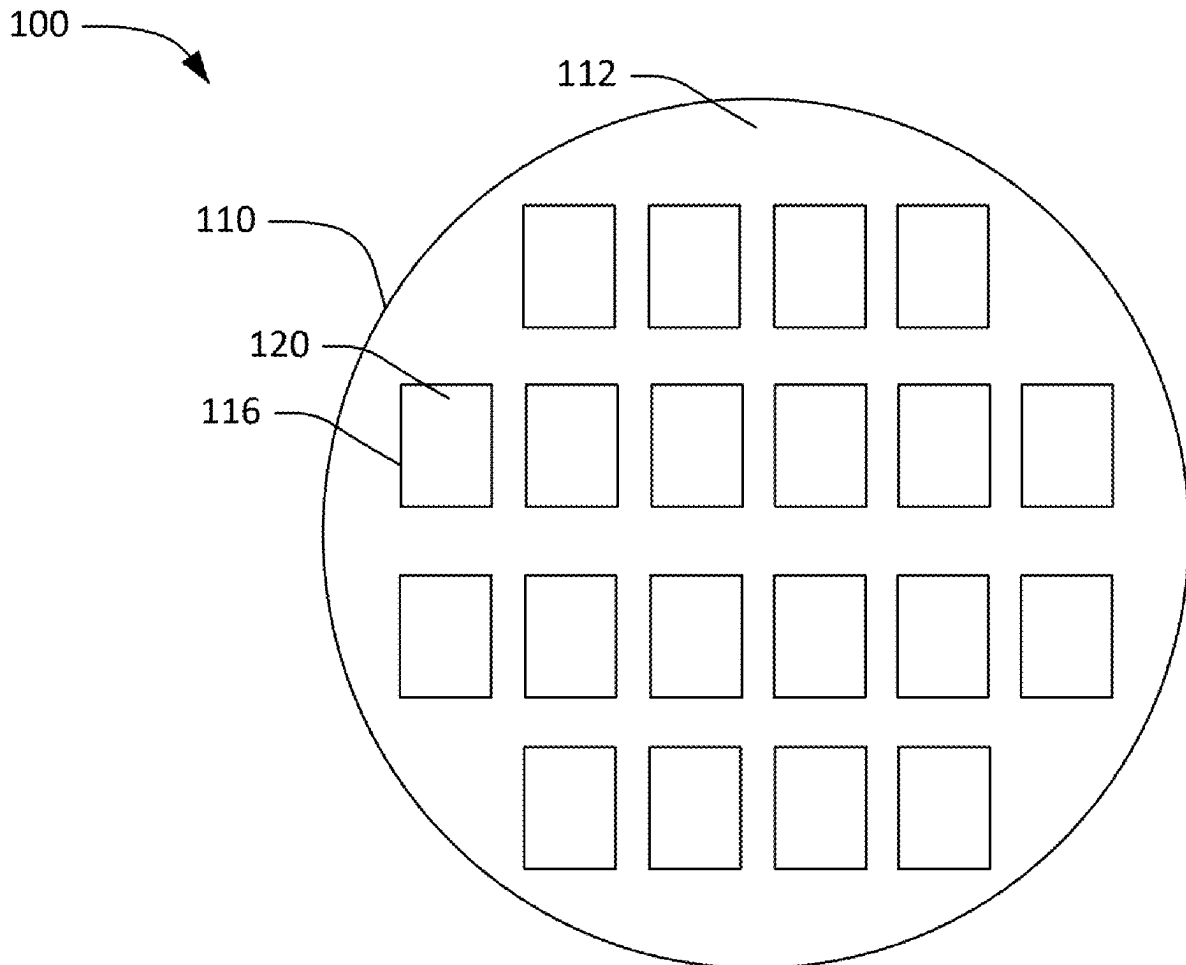
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(57) **ABSTRACT**

An electronic assembly has a wafer having a first circuit
including passive devices for the purpose of one of tuning or
matching networks. Chiplets are placed in the cavities. At
least one chiplet has a second circuit including at least one
transistor or switch device and passive tuning circuits
including at least one of a stabilization network, a gain
boosting network, a power delivery network, or a low-noise
network. Electrical interconnects between the chiplets and
wafer electrically connect the first circuitry to the second
circuitry.



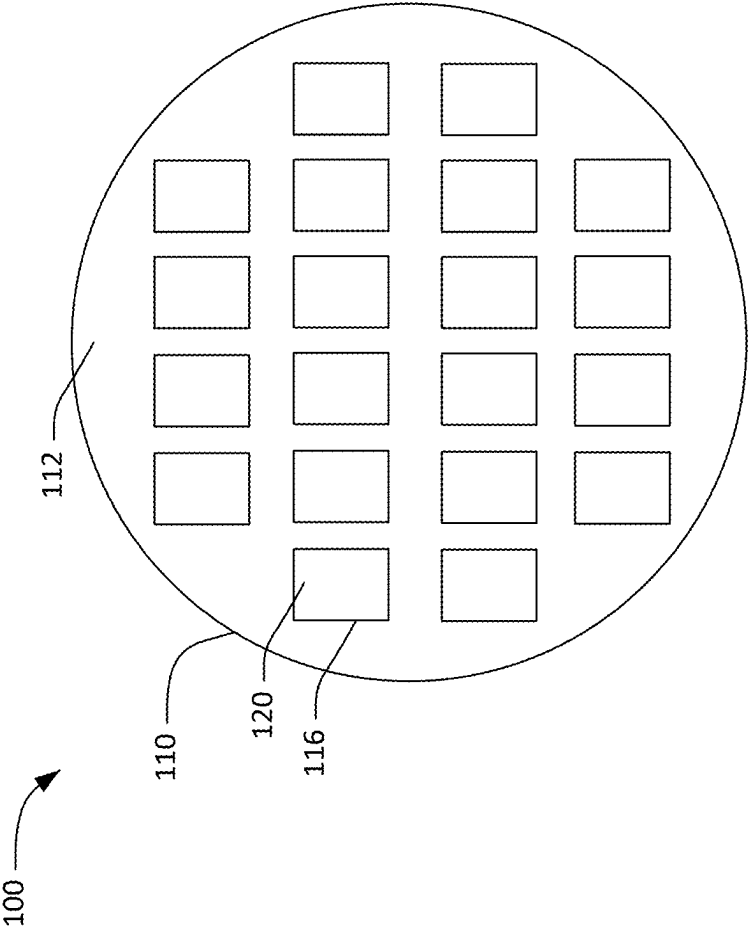


FIG. 1A

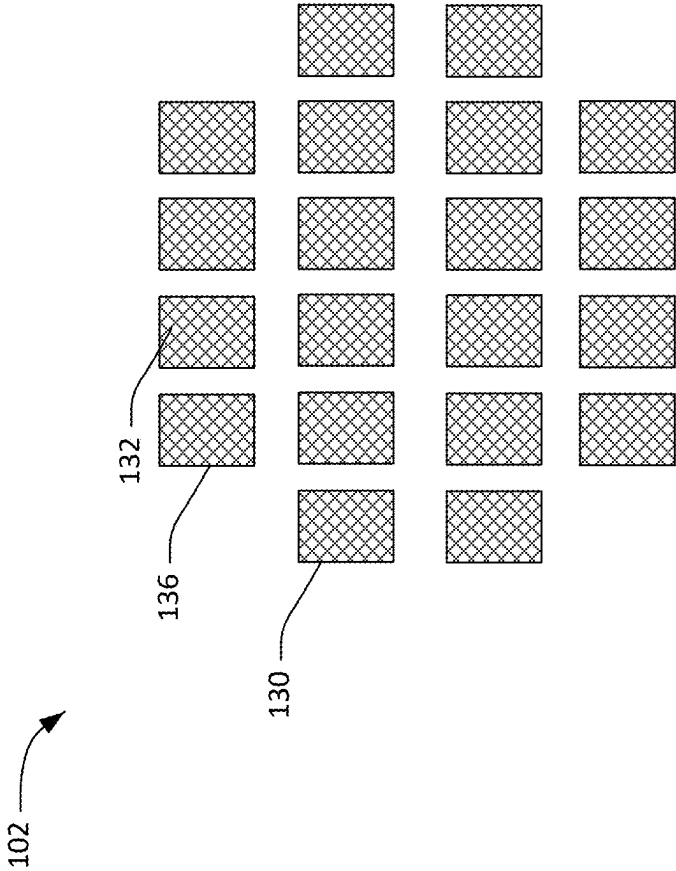


FIG. 1B

106 →

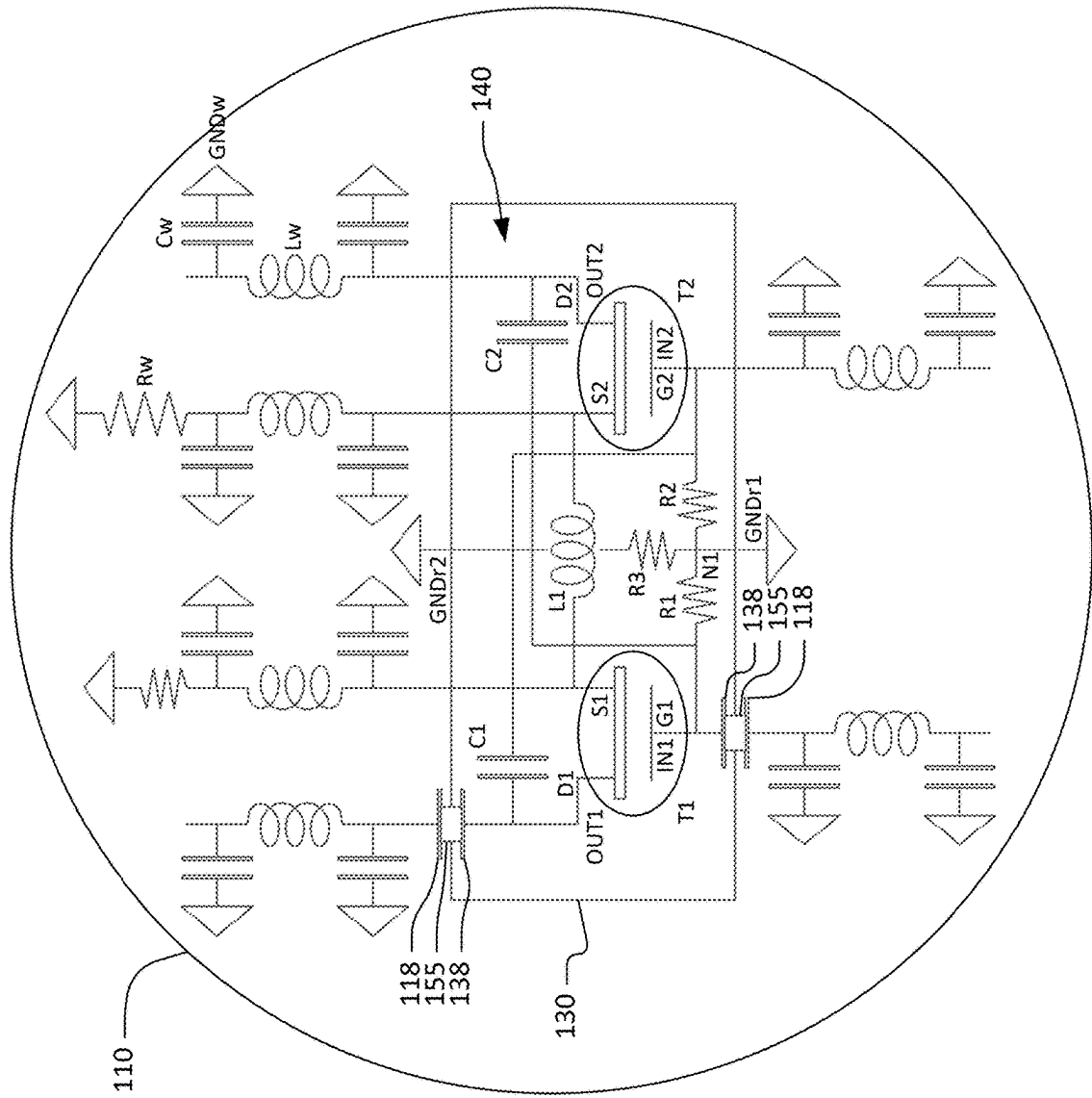
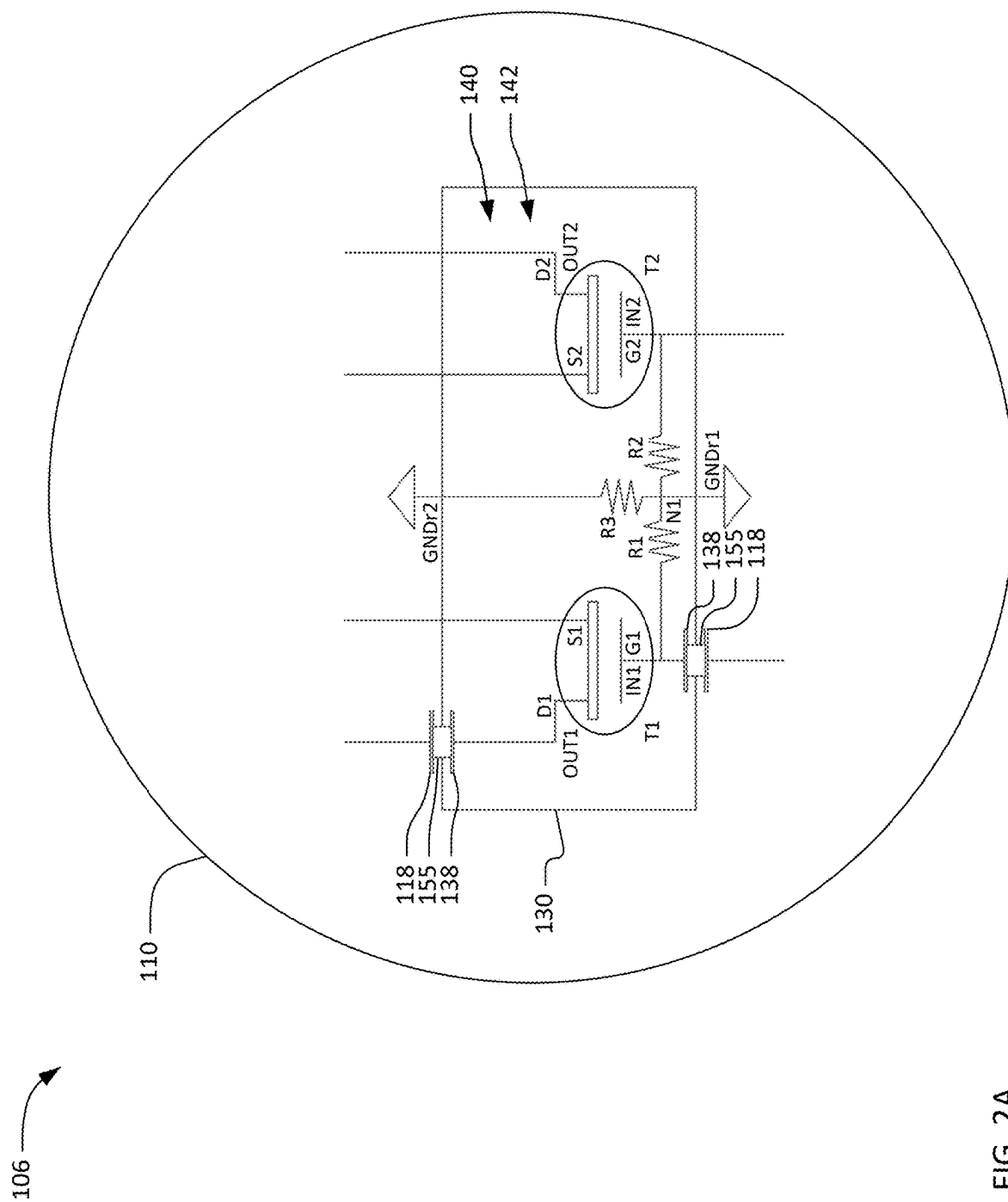


FIG. 1C



106 →

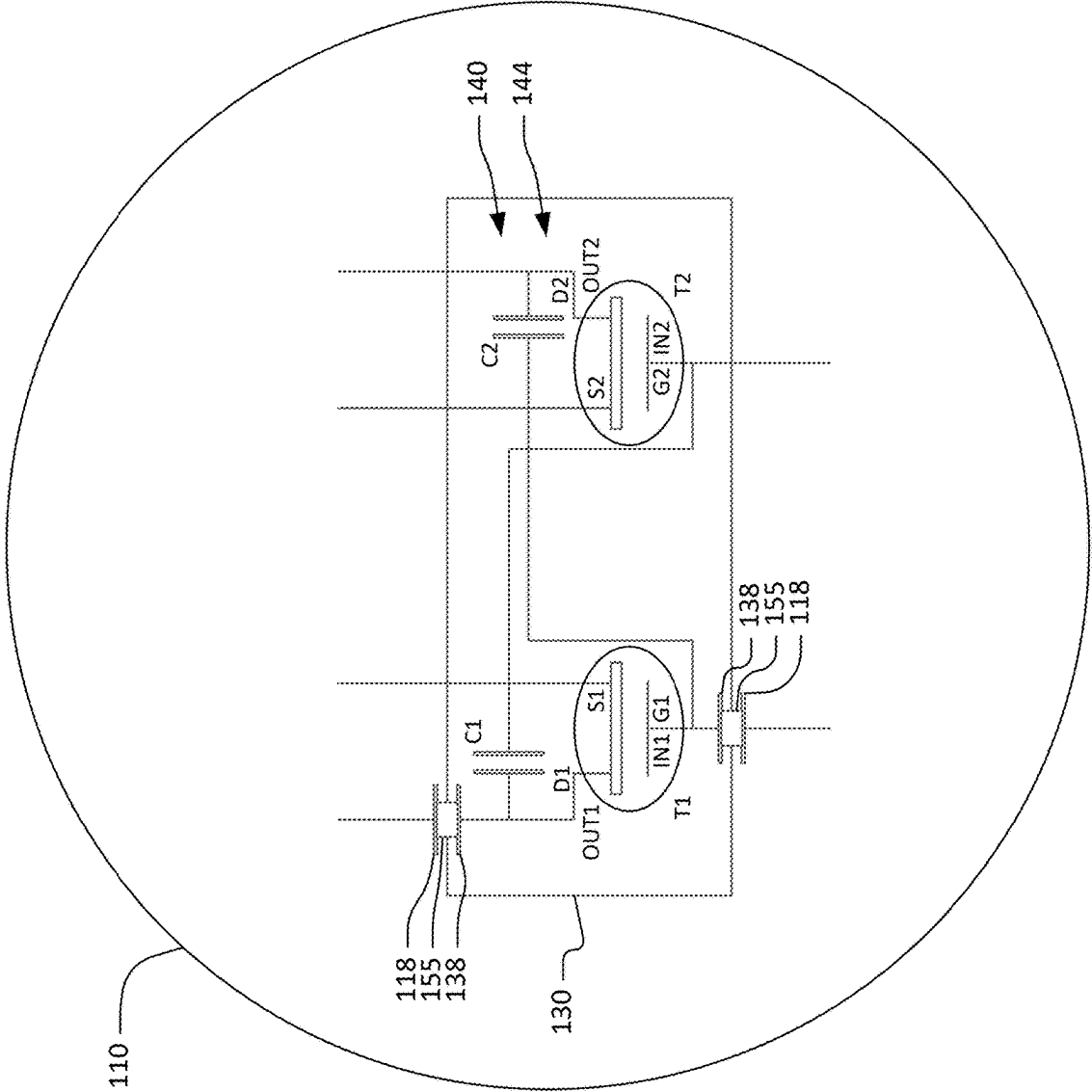
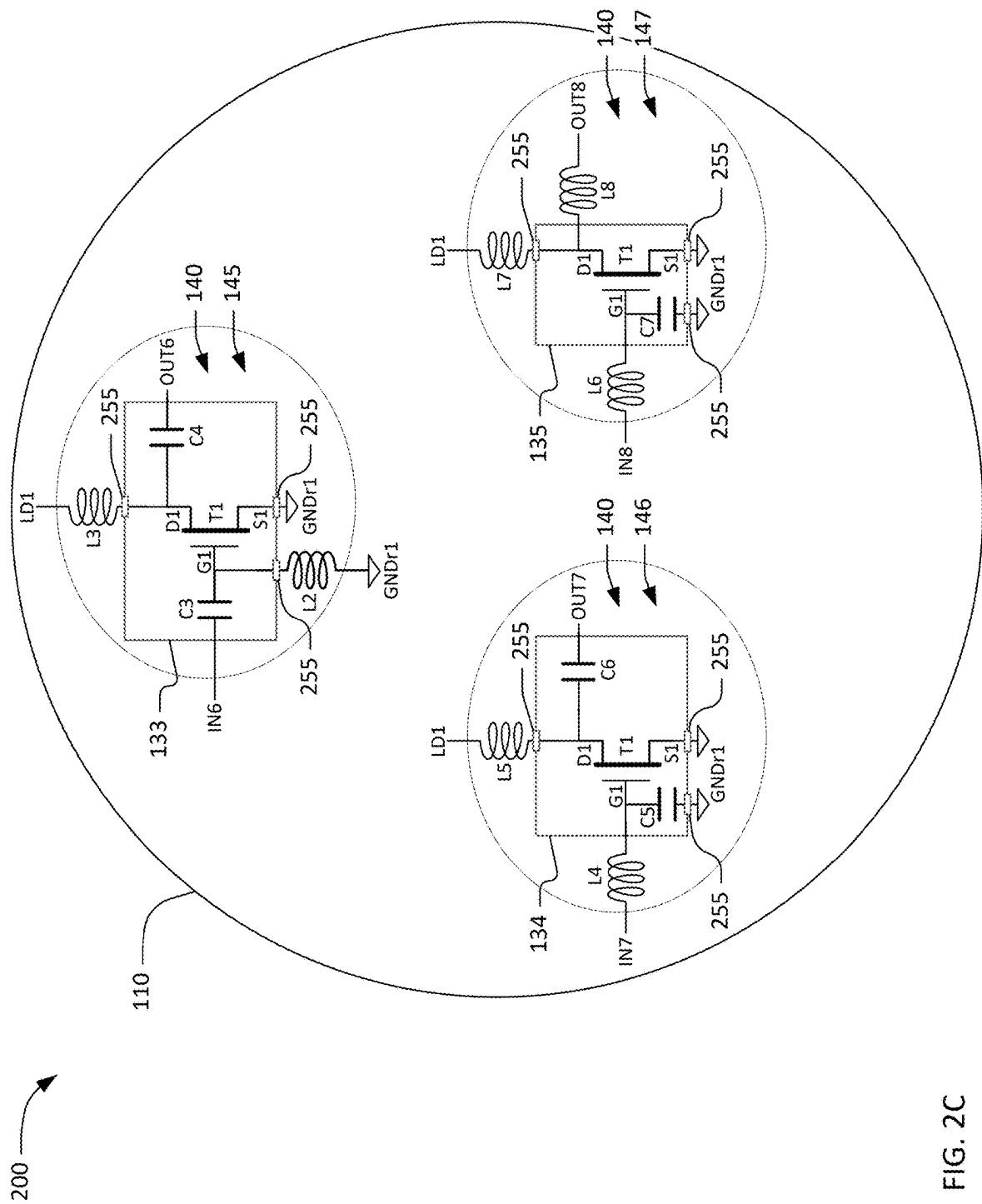


FIG. 2B



106 →

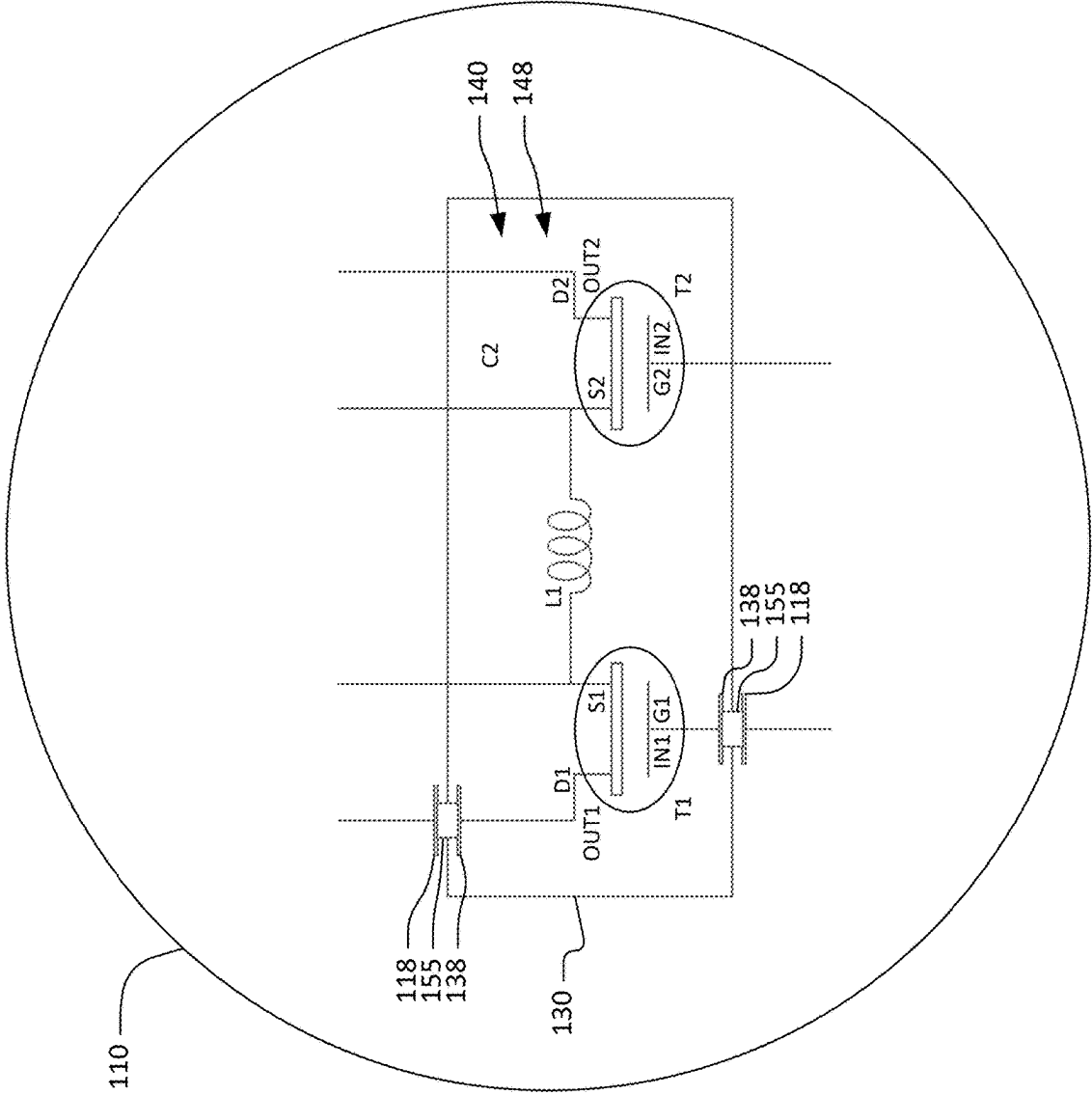


FIG. 2D

250

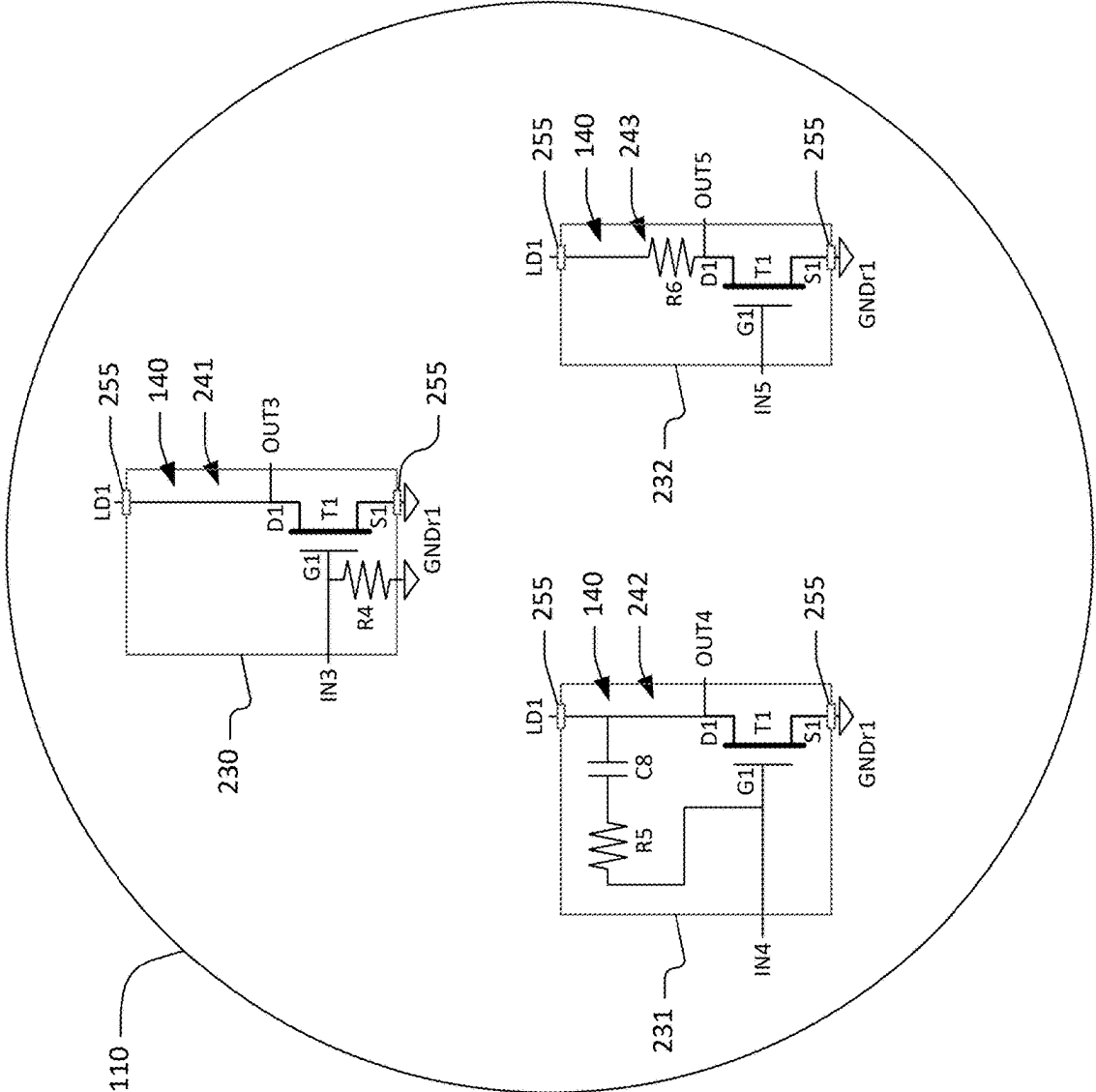


FIG. 2E

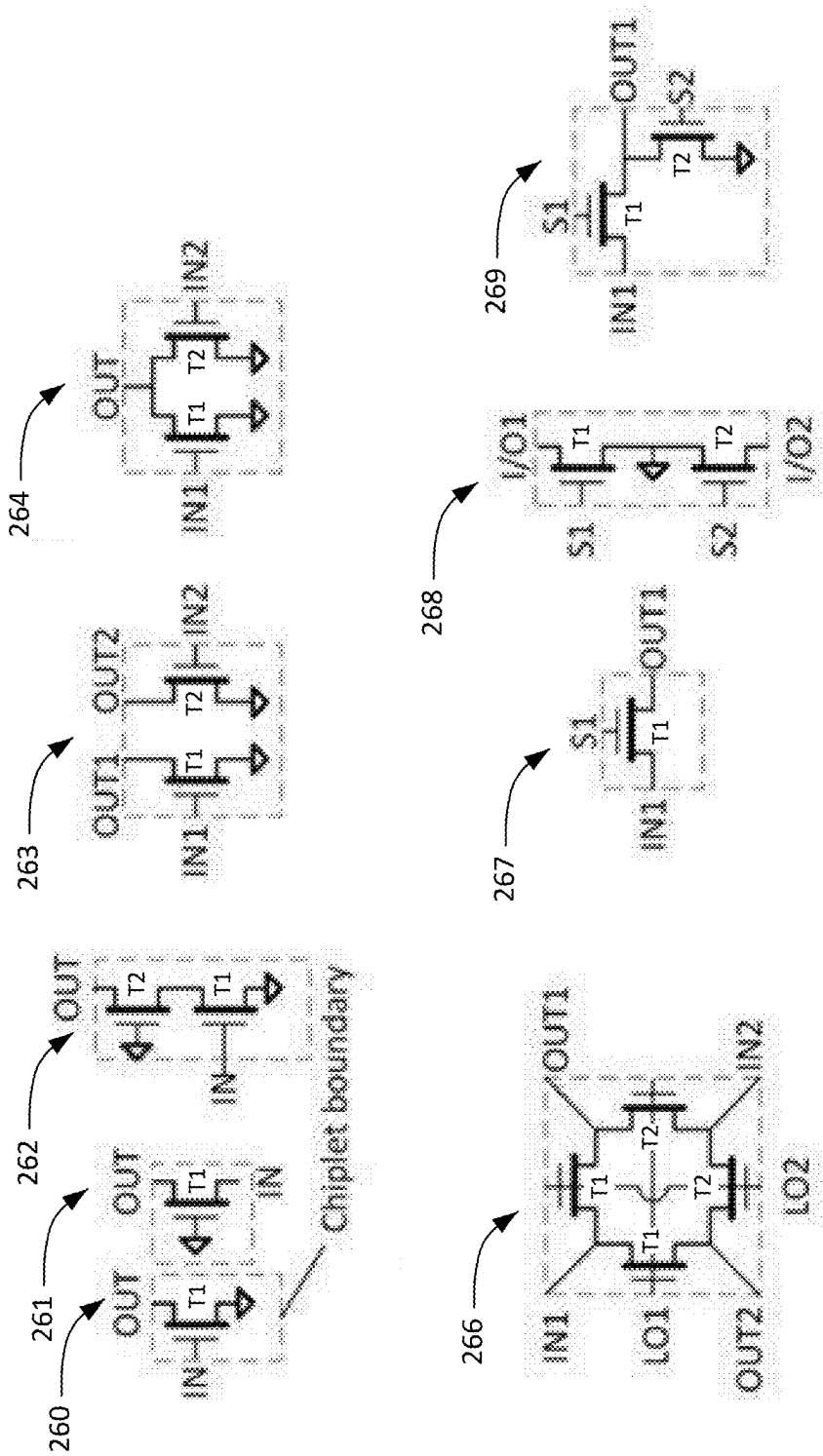


FIG. 2F

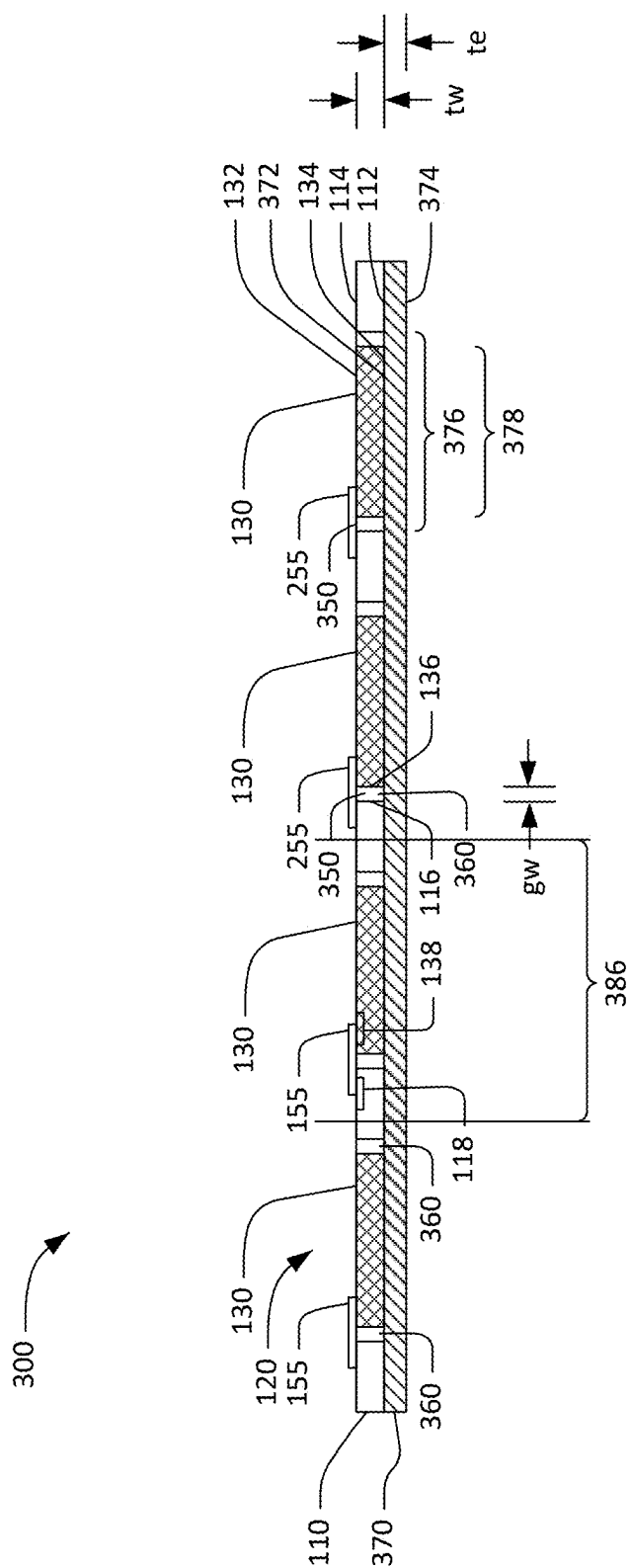


FIG. 3A

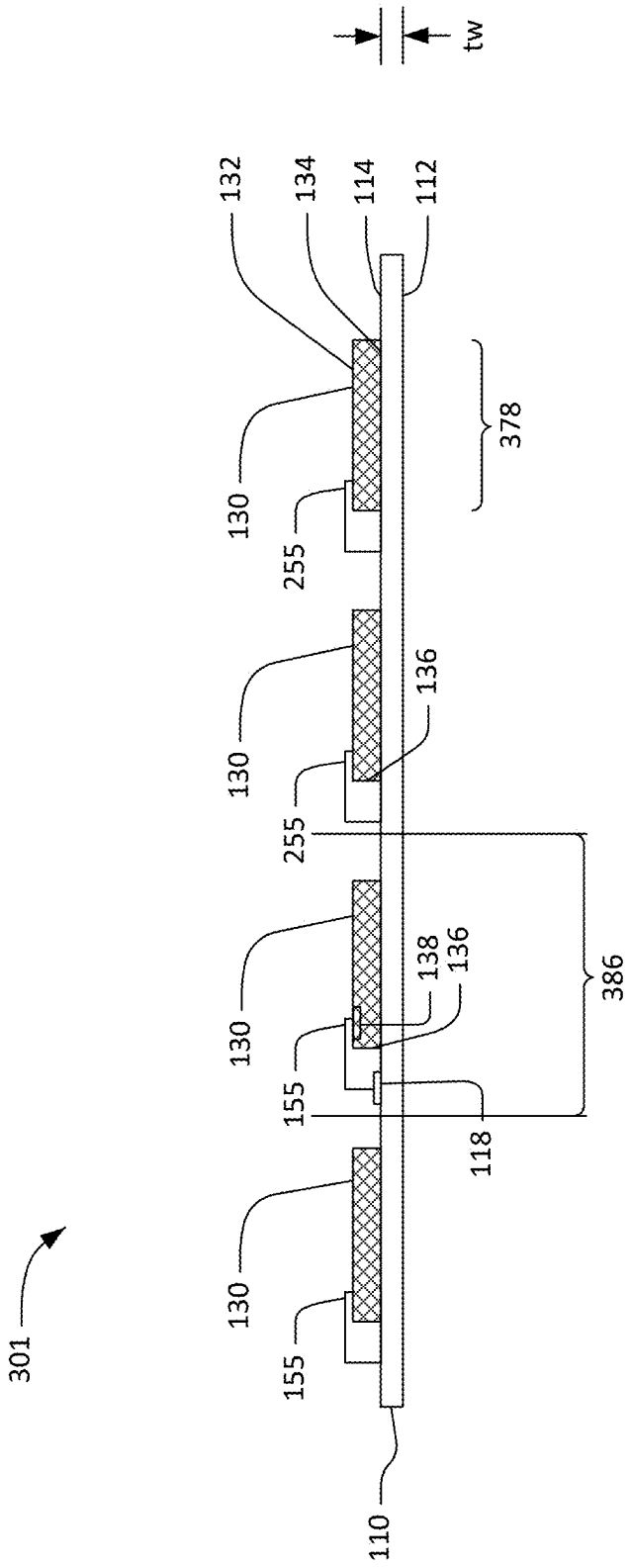


FIG. 3B

HETEROGENEOUS INTEGRATION OF RADIO FREQUENCY TRANSISTOR CHIPLETS HAVING INTERCONNECTION TUNING CIRCUITS

RELATED APPLICATION INFORMATION

[0001] This application is a continuation of co-pending patent application Ser. No. 18/491,661, filed Oct. 20, 2023, which is a continuation of patent application Ser. No. 18/182,314, filed Mar. 10, 2023, issued on Nov. 7, 2023 as U.S. Pat. No. 11,810,876, entitled HETEROGENEOUS INTEGRATION OF RADIO FREQUENCY TRANSISTOR CHIPLETS HAVING INTERCONNECTION TUNING CIRCUITS, all of which is incorporated herein by reference.

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BACKGROUND

Field

[0003] This disclosure relates to wafers, die, chips and fabrication techniques thereof for electronic assemblies having heterogeneous integration of radio frequencies (RF) transistor chips (e.g., chiplets) within wafer cavities of the wafer and having interconnection tuning circuits for tuning interconnections between the wafer and the chips.

Description of the Related Art

[0004] Electronic assemblies, or hybrid circuits, comprise microelectronic circuits fabricated separately and assembled together to form a single component, which can itself be encapsulated in an electronic circuit package. Assembling microelectronic circuits fabricated separately allows, for example, testing of all the microelectronic circuits separately, prior to assembling them, which, in turn enables improved fabrication yields of the final component. This capability is particularly significant if some of the microelectronic circuits fabricated separately are difficult and/or expensive to manufacture. Assembling microelectronic circuits fabricated separately also allows combining of microelectronic circuits, which themselves employ different materials and different manufacturing processes, into a single final component. This capability can lead to higher circuit performance.

[0005] There exists a need for an electronic assembly using a host wafer having pre-fabricated integrated circuitry, such as passive components, that connect to a microelectronics active chiplet (i.e., with transistors) integrated in a through-wafer cavity of the host wafer. This need may for example be for an assembly for microwave or other radio frequency (RF) integrated circuits that decouple the fabrication of the active circuits (e.g., fabrication of the transistors) from the fabrication of the passive circuitry components (e.g., fabrication of the interconnects, resistors and capacitors). Satisfying this need will allow for much faster

manufacturing of the circuits, at lower cost, and a scaling up of active device technologies to circuits without cost and cycle time burden.

DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1A is a schematic top view of a host wafer for heterogeneous integration of radio frequencies (RF) transistor chiplets with tuning circuitry for matching of impedances.

[0007] FIG. 1B is a schematic top view of radio frequencies (RF) transistor chiplets having interconnection tuning circuits for heterogeneous integration into wafer cavities of a host wafer.

[0008] FIG. 1C is a circuit diagram of an electronic assembly having heterogeneous integration of a radio frequencies (RF) transistor chiplet into a wafer, where the chiplet has interconnection tuning circuits.

[0009] FIG. 2A is a schematic diagram of the electronic assembly of FIG. 1C that isolates an example of interconnection tuning circuits having stabilization network circuitry.

[0010] FIG. 2B is a schematic diagram of the electronic assembly of FIG. 1C that isolates an example of interconnection tuning circuits having a neutralization network to improve stability or introduce positive feedback to increase gain of the transistor in the presence of the interconnection to the wafer.

[0011] FIG. 2C is a schematic diagram of an electronic assembly that isolates an example of power delivery to and from chiplets through the interconnection tuning circuits that introduce impedance matching.

[0012] FIG. 2D is a schematic diagram of the electronic assembly of FIG. 1C that isolates an example of interconnection tuning circuits having a negative feedback or degeneration network as is found in low-noise network circuitry to tune the transistor.

[0013] FIG. 2E is a schematic diagram of an electronic assembly that has an example of interconnection tuning circuit that isolates stabilization network circuits of the chiplets in the presence of the interconnect networks.

[0014] FIG. 2F shows circuits having RF transistor chiplets in a wafer, where the chiplets have interconnection tuning circuits.

[0015] FIG. 3A is a schematic cross-sectional view of a device having heterogeneous integration of a radio frequencies (RF) transistor chiplets into a wafer, where the chiplets have interconnection tuning circuits.

[0016] FIG. 3B is a schematic cross-sectional view of a device having heterogeneous integration of a radio frequencies (RF) transistor chiplets onto a wafer, where the chiplets have interconnection tuning circuits.

[0017] Throughout this description, elements appearing in figures are assigned three-digit or four-digit reference designators, where the two least significant digits are specific to the element and the one or two most significant digit may be the figure number where the element is first introduced or fabricated. An element that is not described in conjunction with a figure may be presumed to have the same characteristics and function as a previously-described or subsequently-described element having the same reference designator.

DETAILED DESCRIPTION

Description of Apparatus

[0018] The following describes improved wafers, die, chips and fabrication techniques thereof for electronic assemblies having heterogeneous integration of radio frequencies (RF) transistor chips (e.g., chiplets) within wafer cavities of the wafer and having interconnection tuning circuits for tuning interconnections between the wafer and the chips. The host wafer has circuitry and the radio frequencies (RF) transistor chips within cavities of the wafer, and the chips include one or more types of network circuitry that tune the electrical interconnections between the chips and the wafer. The host wafer can have pre-fabricated interconnects and integrated circuitry, such as passive components, that connect to a chiplet level microelectronics transistor chip integrated in a through-wafer cavity of the wafer. This may form an assembly for integrated circuit devices where the chips contain active circuits from at least one semiconductor technology and the wafers contain passive (or active) circuits from another semiconductor technology (often a cheaper and larger scale technology). Using a low-cost large-diameter integration platform for the chips with active devices allows for much faster manufacturing of the assembled circuits, at larger scale and lower cost.

[0019] The electronic assembled circuit may integrate chiplets having one type of components into a carrier wafer having a different type of components. The electronic assembled circuit may integrate chiplets having high-performance integrated circuits, such as Gallium Nitride (GaN) radio frequency (RF) integrated circuits (ICs) into host wafers having other integrated circuits, such as silicon-based integrated circuits, in a manner that is inexpensive and has high manufacturing yields and short manufacturing cycles. The high performance RF ICs, chips (or chiplets) can have type III-V transistors or other types of transistors and passive circuitry or components, and can be integrated together with resistors, inductors, capacitors and matching networks, as well as active devices from another semiconductor technology into the host wafer. For example, the RF ICs or chiplets can be one type of semiconductor technology that is integrated together with resistors, inductors, capacitors, matching networks, active devices from another semiconductor technology that are part of the host wafer.

[0020] A chiplet may be a chip including the circuitry, material, and/or devices noted herein. It may also be a chip or small chip having active (i.e., transistor) microelectronic devices, active CMOS devices, active microwave IC devices and/or active radio frequency (RF) IC devices. It may also be a chip or small chip having a surface acoustic wave (SAW), bulk acoustic wave (BAW) or other acoustic wave device. A chiplet may have a footprint or top surface area that is half, a third, a fifth or less than a fifth of that of a computer processor chip (e.g., 8086, P3, P4, etc.). Active devices may include active electronic components and/or active electronic circuits.

[0021] The interconnect between the chiplet and wafer maybe mistuned, such as by having AC (alternating current) signal frequencies different at the wafer as compared to the chiplet. The interconnect between the chiplet and wafer maybe mismatched such as by having impedance differences at the wafer as compared to the chiplet that cause instability in AC signals, DC (direct current) signals and/or other electronic signal matching with the input and/or output to

the chiplet or a transistor on the chiplet. This lack of tuning may be improved or tuned using the tuning circuits or networks herein.

[0022] FIG. 1A is a schematic top view **100** of a host wafer **110** for heterogeneous integration of radio frequencies (RF) transistor chiplets with tuning circuitry for matching of impedances. Host wafer **110** has back surface **112** and front surface **114** as shown in FIGS. 3A and 3B. Host wafer **110** and/or each cavity **120** has side surfaces **116**, such as a vertical or sidewall surfaces between the back surface **112** and front surface **114**. There may be 3, 4 or more side surface **116**. Typically, there are 4 side surfaces. Wafer capacitors, inductors and resistors may be tuning circuitry for matching of impedances of the interconnects from the chiplet (e.g., see FIG. 1C).

[0023] Host wafer **110** may be or include (e.g., as a mixture of materials or as material layers) silicon, silicon germanium, silicon on insulator, gallium arsenide, indium phosphide, aluminum nitride, diamond, quartz, alumina. If the host wafer **110** only contains interconnections and passive components, it can be a dielectric such as quartz, alumina, or another ceramic. The host wafer **110** may have layers of one or more of these materials in the form of an oxide material, crystalline material and polycrystalline material and/or amorphous material. Host wafer **110** may include some or only passive components such as resistors, capacitors, inductors, through substrate vias, dielectric layers, and/or metal layers (e.g., signal traces or signal planes). Host wafer **110** may include at least one layer of silicon, silicon carbide (SiC), quartz, or another semiconductor wafer material.

[0024] Host wafer **110** may include areas to be diced into integrated circuits, each having passive integrated components (e.g., signal traces, interconnects and conductive vias, resistors, inductors and/or capacitors), a single transistor and/or a plurality of transistors. Silicon is an advantageous choice for host wafer **110**, because it takes advantage of having a lower expense than other materials; and/or of known microelectronics fabrication processes and of scaling and manufacturing capabilities.

[0025] FIG. 1B is a schematic top view **102** of radio frequencies (RF) transistor chiplets **130** having interconnection tuning circuits for heterogeneous integration into wafer cavities **120** of host wafer **110**. Chiplets **130** have frontside **132** (e.g., a frontside surface) and backside **134** (e.g., a backside surface) as shown in FIGS. 3A and 3B. Each chiplet **130** has side surfaces **136**, such as a vertical or sidewall surfaces between the frontside surface **132** and backside **134**. There may be 3, 4 or more side surface **136**. Typically there are 4 side surfaces. The number of side surfaces **136** of each chiplet **130** may be the same as the number of surfaces **116** of the cavities in the host wafer **110**. Chiplets **130** may each include circuitry described for FIGS. 1C-3B.

[0026] Chiplets **130** may each be or include (e.g., as a mixture of materials or as material layers) silicon (Si), silicon germanium (SiGe), silicon-on-insulator, gallium arsenide (GaAs), indium phosphide (InP), aluminum nitride, quartz, alumina, gallium nitride (GaN). The chiplets **130** may have layers of one or more of these materials in the form of an oxide material, crystalline material and polycrystalline material and/or amorphous material. There may be different electrical component ones or types of chiplets **130** that are manufactured separately from each other. Chiplets

130 can include a GaN, InP or GaAs or any other industry-known electrical component and can be fabricated on a substrate such as Si, SiGe, InP, GaAs, Alumina, or diamond, or any other substrate known in the industry.

[0027] Chiplets **130** or types of chiplets **130** may include transistors used for RF switches, transmit and/or receive circuits; power switches, amplifiers and circuits such as using GaAs, InP, GaN while the host wafer may include transistors such as Si CMOS transistors. The chiplet transistors may have smaller and more expensive electrical components than those of host wafer **110**. There may be one, tens, hundreds, thousands or hundreds of thousands of chiplets **130** embedded in or on one host wafer **110**. There may be between one and one hundred chiplets **130** embedded in one host wafer **110**. Host wafer **110** may have more passive components, lower cost components, routing (e.g., traces, conductive vias and interconnections) than those of chiplets **130**. Host wafer **110** may be fabricated using different microelectronic fabrication techniques or processes than used to fabricate chiplets **130**.

[0028] Chiplets **130** and host wafer **110** can be made of different materials. For example, wafer **110** can be a silicon wafer while chiplets **130** can be a type III-Nitride material component chip. Chiplets **130** may each be or include an integrated circuit having passive integrated components (e.g., signal traces, interconnects and conductive vias, resistors, inductors and/or capacitors), a single transistor and/or a plurality of transistors.

[0029] The chiplets **130**, each include at least one of transistor circuitry and interconnects to contact pads on a frontside **132** of the chiplets **130**. The chiplets **130** may be high-end pre-fabricated active device chiplets that are integrated into host wafer **110** through pick and place assembly on temporary wafer with an adhesive laminate or simply on an adhesive laminate.

[0030] FIG. **1C** is a schematic circuit diagram of an electronic assembly **106** having heterogeneous integration of radio frequencies (RF) transistor chiplet **130** into host wafer **110**, where the chiplet **130** has interconnection tuning circuits **140** having passive circuitry. Wafer **110** includes resistors as shown by their schematic symbol one of which is labeled R_w , capacitors as shown by their schematic symbol one of which is labeled C_w , inductors as shown by their schematic symbol one of which is labeled L_w , and grounding as shown by their schematic symbol one of which is labeled GND_w . In some cases, the wafer capacitors and inductors, such as shown by their CLC network under resistor R_w , represent the electrical characteristics of an interconnect such as interconnect **155** and/or **255**. Here, the CLC network may represent an interconnect between the chiplet and the wafer. These characteristics may be part of what is being tuned and/or be part of the interconnection tuning circuitry on the chiplet and/or wafer. The resistor R_w might be considered a separate interconnection tuning circuit on the wafer **110**. The wafer capacitors, inductors and resistors may be tuning circuitry for matching of impedances of the interconnects from the chiplet. The wafer circuitry may be described as passive (e.g., R, C, L, non-transistor) devices for the purpose of one of tuning or matching networks. Other passive devices may be in or on the wafer. In some cases, the wafer **110** only includes passive, non-transistor devices. In other cases, the wafer circuitry may also include active devices that are not radio frequencies (RF) transistor devices.

[0031] Chiplet **130** includes resistors R_1 , R_2 and R_3 ; capacitors C_1 and C_2 ; inductor L_1 , Transistors T_1 and T_2 . Transistor T_1 has source S_1 , drain D_1 which is considered the transistor output OUT_1 and gate G_1 which is considered the transistor input IN_1 . Transistor T_2 has source S_2 , drain D_2 which is considered the transistor output OUT_2 and gate G_2 which is considered the transistor input IN_2 . The chiplet circuitry may be described as having both passive (e.g., R, C, L) and active (e.g., at least transistors T_1 and T_2) devices. Other passive and/or active devices may be in or on the chiplet.

[0032] In some cases, an input signal to the transistor T_1 and/or transistor T_2 ; or an input signal to the chiplet such as input IN_x (e.g., where x is a number as indicated herein); or another input signal to the chiplet also includes a second ground signal, such as GND_{r1} or GND_{r2} or another ground signal from the chiplet to the transistor or wafer **110**.

[0033] A differential input port between G_1/IN_1 and G_2/IN_2 ; and a differential output port between D_1/OUT_1 and D_2/OUT_2 may be used to increase the power generated in a given chiplet area. In this case, S_{22} may be the “reflection” of output power when Port 2 or OUT_2 is connected to a load (not shown). In some cases, an output signal from the transistor T_1 and/or transistor T_2 ; or an output signal from the chiplet such as input OUT_x (e.g., where x is a number as indicated herein); or another output signal from the chiplet also includes a second ground signal, such as GND_{r1} or GND_{r2} or another ground signal to the chiplet or from the transistor or wafer **110**. Here, an input to a transistor or chiplet may have a port 1 between the gate of the transistor and GND_{r1} ; and have an output from the transistor between the drain of the transistor and GND_{r1} . The input port 1 may have a reflection of input power from a source that is characterized by S_{11} ; the output port 2 may have a reflection of output power at the load (e.g., LD_1) that is characterized by S_{22} ; a forward propagation (gain) of the wave from port 1 to port 2 that is S_{21} ; and a reverse propagation (isolation) of the wave from port 2 to port 1 that is S_{12} .

[0034] The available gain G_A of the S-parameter network may be characterized by S parameters and the source impedance as:

$$G_A = \frac{|S_{21}|^2(1 - |\Gamma_S|^2)}{|1 - S_{11}\Gamma_S|^2 - |S_{22} - \Delta\Gamma_S|^2}$$

[0035] The unilateral gain U may be invariant to a lossless embedding network of any of the networks **140** as:

$$U = \frac{|S_{12} - S_{21}|^2}{\det[1 - SS^*]}.$$

[0036] These S-parameters are useful for characterizing how stable a circuit is. A family of stability circles exists over all frequency and input powers. The circuit is only unconditionally stable if the μ_u factor remains greater than 1 for all stability circles possible operating conditions, such as where:

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12} S_{21}|} > 1.$$

[0037] That is any of the networks **140**, without circuitry of the wafer **110** (except with the network of FIG. 2C of inductors L3-L8 on the wafer) may make the transistor or chiplet unconditionally stable by causing μ to be greater than 1 under operating conditions, such as when the transistor or chiplet is amplifying or passing an RF signal.

[0038] Each of transistors T1 and T2 may be a radio frequencies (RF) transistor device. Each of transistors T1 and T2 may operate at (e.g., amplify, pass and/or switch) radio frequencies and/or be an RF wave transistor device. Each may operate on electronic signals with frequencies between 10 MHz and 300 GHz, 30 GHz and 300 GHz or 10 MHz and 10 GHz. Each of transistors T1 and T2 may be a millimeter-wave or a microwave transistor device. Each of transistors T1 and T2 may be a single ended RF amplifier, such as used to make a chain of amplifiers, a receiver using amplifiers, a mixer, a switch or a frequency multiplier. Each of transistors T1 and T2 may be part of a differential amplifier, placed in parallel with a shared drain and source but distinct gate or base connections for a frequency multiplier, or in a series connection of devices for a cascode. Additionally, transistors T1 and/or T2 might be part of a switch pair of transistors for an RF switch or mixer. Four total transistors might be used in a single chiplet for a FET ring mixer. Each of transistors T1 and T2 may be a heterojunction bipolar transistor (HBT), in which case they will have a base, emitter and collector instead of a gate, source and drain, respectively. Each of transistors T1 and T2 may be used in common gate mode by grounding both transistor's gates and using the sources as inputs. Each of transistors T1 and T2 may be used with the input at the source and an output at the drain.

[0039] In the example of FIG. 1C, transistor T1 is an RF amplifier and has a signal phase output of a differential amplifier, and transistor T2 is another RF amplifier and has a signal phase output of the differential amplifier that is 180 degrees out of phase with the signal phase of transistor T1. Transistors T1 and T2 may be two differential RF transistors and there may be interconnects from these transistors to contact pads **138** on a frontside surface of the chiplet **130**.

[0040] Wafer **110** may also include circuit networks to provide a virtual ground at GNDr1 such as an AC ground and circuitry to apply the DC bias voltage applied to the node N1 between resistors R1 and R2. Wafer **110** may also include GNDr2 such as an AC ground and DC bias voltage applied to one end of resistor R3 that is opposite the node N1 between resistors R1 and R2.

[0041] Assembly **106** also includes interconnects **155** between contacts **118** of wafer **110** and contacts **138** of chiplet **130**. A contact may be a contact pad or other electrical connection to circuitry of the wafer or chiplet. The interconnect **155** may be a trace (e.g., of a PCB), a wire or another electrically conductive connection between the contacts. The electrical interconnects **155** electrically connect the wafer circuitry to the chiplet circuitry. There may be similar interconnects and contact pair at the border (e.g., at sidewalls) between the wafer and chiplet for each electrical connection needed at that border.

[0042] The electrical interconnects **155** electrically connect the circuit devices of the wafer and the transistors of the

chiplets. The interconnect between the chiplet and wafer maybe mistuned, such as by having mismatches in stability, AC signal, DC signal, impedance, frequency and/or other electronic signal matching with the input and/or output to the chiplet or a transistor on the chiplet. This lack of tuning may be improved or tuned using the tuning circuits **140**. Notably, the interconnection tuning circuits **140** reduce this electrical mistuning of passive devices and the transistors caused by the electrical interconnects that may impact circuit performance and reduces the uncertainty of the electrical impedance and frequency of signals passed across the interconnects to improve the circuit performance.

[0043] The interconnection tuning circuits **140** may be any or all of the resistors, inductors and capacitors shown on the chiplet **130** or other chiplets herein. In some cases, it may also include inductors on the wafer such as shown in FIG. 2C. The interconnection tuning circuits **140** may include at least one of a stabilization network (e.g., network of electronic circuitry), a gain boosting network, a power delivery network, and/or a low-noise network. Each of these networks may be an embedded network that is embedded into wafer **110** as part of the chiplet.

[0044] There may be more or fewer electrical and/or circuit components than shown in FIG. 1C for chiplet **130**, wafer **110**, and/or interconnection tuning circuits **140**. Although only one is shown in FIG. 1C, there may be multiple chiplets **130** in cavities of wafer **110**. There may be between 10 and 1000 such chiplets in the wafer.

[0045] FIG. 2A is a schematic diagram of the electronic assembly **106** of FIG. 1C that isolates an example of stabilization network circuitry **142** of the interconnection tuning circuits **140**. Circuitry **142** may be a network of electronic circuitry having passive circuitry that includes at least one of resistors R1, R2 and/or R3. Circuitry **142** may include only resistors R1 and R2. In another case, circuitry **142** includes all of resistors R1, R2 and R3. The R3 resistor is introduced to provide common-mode stability or to provide feedback to the virtual ground on the drain/output circuitry. Circuitry **142** may optionally include ground GNDr1 and/or GNDr2. Circuitry **142** may optionally include one or more of inductor L1, capacitor C1 and/or capacitor C2. The stabilization network circuitry **142** may be stabilization network circuit for transistor T1 and/or T2 in spite of the electrical characteristic or mistuning of interconnection circuit **155** and/or **255**. Additionally, the resistor R3 may not grounded to GNDr1 but may instead connect to the drain D1 and/or D2 of the transistor T1 and/or T2 through some external circuit that exists in the wafer. In some cases, resistors R1, R2 and R3 are not grounded to ground GNDr1 or GNDr2 (although they may be grounded to a "virtual" ground), and are used to provide a DC bias to the gates G1 and G2 while not influencing the differential circuit's RF frequency performance. The ground GNDr1 or GNDr2 may provide DC bias voltage through resistors R1 R2 and R3 to the gates G1 and G2. Thus, the resistors R1 R2 and R3 provide stabilization of the DC bias to the gates G1 and G2, such as by setting the DC bias on the gates of the transistors.

[0046] In some cases, circuitry **142** may include at least one resistor R1 between the input IN1 of transistor T1 and the node N1 to ground GNDr1; at least one resistor R2 between the input IN2 of transistor T2 and the node N1; and at least one resistor R3 between the node N1 and ground GNDr2. Here, ground GNDr1 may be a first AC ground that allows a DC voltage that provides a desired first DC bias to

the inputs IN1 and IN2; and GNDr2 may be a second AC ground and DC voltage that provides a desired second DC bias to the inputs the inputs IN1 and IN2.

[0047] In some cases, circuitry 142 may include at least one resistor R1 (and optionally R2) between the input IN1 and the input IN2; and at least one resistor R2 (and optionally R1) between the input IN2 and the input IN1, as shown in FIG. 1C. There may optionally be capacitors C1 and C2 between these inputs and outputs as shown in FIG. 1C. There may optionally be inductor L1 between the S1 and S2 as shown in FIG. 1C.

[0048] Any one, two or all 3 of these resistors R1, R2 and R3 may provide desired DC bias and common mode stability to the inputs IN1 and IN2 as compared to without the resistor(s). Any one, two or all 3 of these resistors R1, R2 and R3 may provide desired impedance matching at the input and output of each transistor T1 and T2 as compared to without the resistor(s).

[0049] In some cases, these resistors R1 and R2 may provide desired differential mode (e.g., where the operation or output of transistor T2 is 180 degrees out of phase with transistor T1) stability to the inputs IN1 and IN2 as compared to without those resistors; and may provide desired impedance matching at the input and output of each transistor T1 and T2 as compared to without those resistors. This stability with reduce unwanted changes in power levels, output of noise and/or oscillation of the transistor outputs. Here, resistors R1 and R2 may provide desired shunt resistance to ground (e.g., GNDr1) that is used for the stabilization.

[0050] In some cases, the resistor R3 (and optionally a capacitor such as C1 in series with resistor R3) may provide desired common mode (e.g., where the operation or output of transistor T2 is 0 degrees out of or in phase with transistor T1) stability to the inputs IN1 and IN2 as compared to without those resistors; and may provide desired common mode impedance matching at the input and output of each transistor T1 and T2 as compared to without those resistors. This stability with reduce unwanted changes in power levels, output of noise and/or oscillation of the transistor outputs. Here, resistor R3 may provide desired series feedback (e.g., to components of the chiplet or wafer electrically connected to the transistor output) that is used for the common mode stabilization.

[0051] The DC stability and/or impedance matching produces an unconditionally stable chiplet with respect to signal received on the interconnects 155 from the wafer and thus allows the active device transistors T1 and T2 of chiplet 130 to also be unconditionally stable. It is considered that stabilization of the transistors T1 and T2 might incorporate lossy components such as resistors R1, R2 and R3 and reactive components such as capacitors C1 and C2 and inductor L1 to stabilize the transistors T1 and T2 in a specific frequency band of operation as noted for the transistors. The stabilization can be applied to a single transistor but also to a differential pair example of transistors T1 and T2. In this case, both the stability of the differential and common-modes would be guaranteed through the network circuitry 142 implemented on the active chiplet.

[0052] FIG. 2B is a schematic diagram of the electronic assembly 106 of FIG. 1C that isolates an example of gain boosting network circuitry 144 of the interconnection tuning circuits 140. Circuitry 144 may be a network of electronic circuitry having passive circuitry that includes at least one of

capacitors C1 and/or C2. Circuitry 144 may include both of capacitors C1 and C2. Circuitry 144 may optionally include ground GNDr1 and/or GNDr2. Circuitry 142 may optionally include one or more of resistors R1, R2 and/or R3, such as shown in FIG. 1C. Circuitry 144 may be a neutralization network to improve stability of transistor T1 and/or T2; or to introduce positive feedback to transistor T1 and/or T2 to increase gain of the transistors in the presence of or through the interconnections 155 and/or 255 to the wafer.

[0053] In some cases, circuitry 144 may include at least one capacitor C1 between the input IN2 of transistor T2 and the output OUT1 of transistor T1; and at least one capacitor C2 between the input IN1 of transistor T1 and the output OUT2 of transistor T2. These capacitors counter-act the parasitic gate-drain capacitance of transistors T1 and T2.

[0054] In some cases, circuitry 144 may include at least one resistor R1 and R2 between the input IN1 and input IN2 as shown in FIG. 1C. There may optionally be inductor L1 between the S1 and S2 as shown in FIG. 1C.

[0055] Both of these capacitors C1 and C2 may provide desired gain boosting and/or a “neutralization”—such as electrically isolating the input IN from the output OUT of each transistor T1 and T2 of the chiplet as compared to without the capacitor(s). Any one or both of these capacitors C1 and C2 may provide desired impedance matching of the impedance the transistor sees at its input IN and output OUT from the wafer 110 as compared to without the capacitor(s). In some cases, the network circuit 144 implemented on the active chiplet 130 can realize desired specific optimization for gain of the transistors T1 and T2. For instance, both of these capacitors C1 and C2 may provide desired unilateralization to improve the gain of the transistor(s) by eliminating the reverse signal flow of the transistor(s) (e.g., cause $S_{12}=0$) as compared to without the capacitor(s). In other words, each of transistors T1 and T2 can be unilateralized such that the reverse isolation (S_{12} —backwards transfer function from OUT to IN) is zero. Any one or both of these cross-coupled capacitors C1 and C2 may provide a desired negative capacitance to cancel C_{gd} (the capacitance between the gate and drain) of the transistor(s) as compared to without the capacitor(s). When operating in differential mode, where transistor T2 is 180 degrees out of phase with transistor T1, the cross coupled capacitors C1 and C2 act as a negative capacitor to the transistors T1 and T2, thus canceling the C_{gd} of each respective transistor as compared to without the capacitors. Another gain boosting technique or circuit (not shown) can be used to increase the gain beyond the unilateral gain. The limit of either or both of these gain boostings, while remaining unconditionally stable, is to reach $4U$ (e.g., 4x the maximum unilateral gain of the T1 and T2).

[0056] In other embodiments, any one or both of these capacitors C1 and C2 may be replaced with an inductor such as an inductor as noted herein between the gate of one of transistor T1 and T2 and the drain of the other one of one of transistor T1 and T2.

[0057] FIG. 2C is a schematic top view of an electronic assembly 200 that isolates an example of power delivery circuits 145, 146 and 147 of the interconnection tuning circuits 140 having passive circuitry of chiplets 133, 134 and 135, each having active transistor T1. In some cases, one of power delivery and efficiency circuits 145, 146 or 147 is also on the chiplet having active transistor T2, such as where chiplets 133, 134 and 135 have both transistor T1 as shown

in FIG. 2C and also have transistor T2 (e.g., see assembly 106) with one of circuits 133, 134 or 135. Each of circuits 145, 146 and 147 may be an example of a power delivery circuit to and from chiplets 133, 134 and 135 through the interconnection networks including through interconnections 155 and/or 255 that introduce impedance matching through interconnections 155 and/or 255, such as from the wafer to transistor T1.

[0058] There may be more or fewer electrical and/or circuit components than shown in for chiplets 133, 134 and 135, wafer 110, and/or network circuits 133, 134 or 135. The circuitry of chiplets 133, 134 and 135 may be described as having both passive (e.g., R, C, L) and active (e.g., at least transistor T1) devices. Other passive and/or active devices may be in or on the chiplet. Although only three chiplets are shown, there may be numerous chiplets in cavities of wafer 110. There may be between 10 and 1000 such chiplets in the wafer.

[0059] Each of chiplets 133, 134 and 135 may represent a version of chiplet 130. Each of circuits 145, 146 or 147 may represent a version of circuitry 140.

[0060] Interconnects 255 are examples of interconnects such as those described by contact 118, interconnect 155 and contact 138. The electrical interconnects 255 electrically connect the wafer circuitry to the chiplet circuitry. There may be interconnects similar to interconnects 255 at the border (e.g., at sidewalls) between the wafer and chiplet for each electrical connection needed at that border.

[0061] In some cases, circuitry 145, 146 or 147 may include at least one capacitor C5 or C7 between the gate G1 of transistor T1 and the ground GNDr1; and/or at least one capacitor C4 or C6 between the drain D1 of transistor T1 and the output OUT6 or OUT 7 of chiplets 133 and 134.

[0062] Circuitry 145 may be a network of electronic circuitry that includes at least two capacitors C3 and C4 and two inductors L2 and L3. Inductor L2 is connected between gate G1 and ground GNDr1 and/or other active or passive circuitry of the wafer 110. Inductor L3 is connected between the load LD1 and the node between drain D1 and capacitor C4. Capacitor C3 is connected between the input IN6 and the node between the inductor L2 and gate G1. Capacitor C4 is connected between the drain D1 and the output OUT6. Inductor L2 may be used to provide DC bias to gate G1 and inductor L3 may be used to provide DC bias to drain D1. Capacitors C3 and C4 may be used to isolate the transistor from another part of the circuit and/or prevent DC current from flowing into a termination port.

[0063] Load LD1 may be a 50-Ohm load which is a convention used in the definition of S-parameters that describe power flow into and out of a transistor.

[0064] In the case shown, inductors L2 and L3 are on wafer 110, such as due to their larger size than the capacitors. In other cases, inductors L2 and L3 are on chiplet 133.

[0065] Inductors L2 and L3; and capacitors C3 and C4 may create an electrical network that increases power delivery and efficiency for transistor T1 (e.g., at output OUT6), by transforming the impedance of the source and/or load to an impedance seen at the transistor that maximizes the power delivered or the power added efficiency, as compared to without the inductors and capacitors. As an impedance transformation network, any of circuits 145, 146 and 147, is an electrical network that takes one impedance (Z1) at one port (such as the input IN or output OUT of that circuit) and transforms it to another impedance (Z2) at a second port

(such as the other of the input IN or output OUT of that circuit). For transistor T1, they may maximize output power (e.g., at output OUT6) under mismatches in the load impedance (e.g., of LD1) or introduce load modulation for power efficiency under different output power conditions, as compared to without the inductors and capacitors.

[0066] Circuitry 146 may be a network of electronic circuitry that includes at least two capacitors C5 and C6 and two inductors L4 and L5. Capacitor C5 is connected between gate G1 and ground GNDr1. Inductor L5 is connected between the load LD1 and the node between drain D1 and capacitor C6. Inductor L4 is connected between the input IN7 and the node between the capacitor C5 and gate G1. Capacitor C6 is connected between the drain D1 and the output OUT7. Inductor L4 may be used to provide DC bias to gate G1 and inductor L5 may be used to provide DC bias to drain D1. Capacitors C5 and C6 may be used to isolate the transistor from another part of the circuit and/or prevent DC current from flowing into a termination port.

[0067] In the case shown, inductors L4 and L5 are on wafer 110, such as due to their larger size than the capacitors. In other cases, inductors L4 and L5 are on chiplet 134.

[0068] Inductors L4 and L5; and capacitors C5 and C6 may provide desired increased power delivery and efficiency for transistor T1 (e.g., at output OUT7) through impedance transformations, as compared to without the inductors and capacitors. For transistor T1, they may maximize output power (e.g., at output OUT7) under mismatches in the load impedance (e.g., of LD1) or introduce load modulation for power efficiency under different output power conditions, as compared to without the inductors and capacitors.

[0069] Circuitry 147 may be a network of electronic circuitry that includes at least one capacitor C7 and three inductors L6, L7 and L8. In some cases, capacitor C7 is on the host wafer 110 and not within chiplet 135. Inductor L7 is connected between the load LD1 and the node between drain D1 and inductor L8. Inductor L6 is connected between the input IN8 and the node between the capacitor C7 and gate G1. Inductor L8 is connected between the drain D1 and the output OUT8.

[0070] In the case shown, inductors L6-8 are on wafer 110, such as due to their larger size than the capacitors. In other cases, inductors L6-8 are on chiplet 135.

[0071] Inductors L6, L7 and L8; and capacitor C7 may provide desired tuning that increases power delivery and efficiency for transistor T1 (e.g., at output OUT8) through impedance transformations, as compared to without the inductors and capacitors. For transistor T1, they may maximize output power (e.g., at output OUT8) under mismatches in the load impedance (e.g., of LD1) or introduce load modulation for power efficiency under different output power conditions, as compared to without the inductors and capacitors. By moving the inductors to the host wafer 110, substantial reduction in cost and time is achieved. In some cases, it is less expensive and results in fewer defects when the inductors are formed on the wafer 110 instead of on a chiplet because the wafer is a cheaper and easier to process semiconductor material.

[0072] Power delivery and efficiency circuits 145, 146 and 147 may also be used on either or both of transistors T1 and/or T2 as shown in FIG. 1C.

[0073] FIG. 2D is a schematic top view of the electronic assembly 106 of FIG. 1C that isolates an example of low-noise network circuitry 148 of the interconnection tun-

ing circuits 140. Circuitry 148 may be a network of electronic circuitry having passive circuitry that includes at least one inductor L1. Circuitry 148 may include only inductor L1. In another case, circuitry 148 includes all of inductor L1 and capacitor C1 and/or capacitor C2. Circuitry 148 may be an interconnection tuning circuit having negative feedback or be a degeneration network through interconnects 155 and/or 255 as is found in low-noise network circuitry to tune the transistor T1 and/or T2.

[0074] In some cases, circuitry 148 may include at least one inductor L1 between the source S1 of transistor T1 and the source S2 of transistor T2. This inductor may have a value between 5 pH and 500 pH.

[0075] In some cases, circuitry 148 may also include at least one capacitor C1 between the input IN2 of transistor T2 and the output OUT1 of transistor T1; and at least one capacitor C2 between the input IN1 of transistor T1 and the output OUT2 of transistor T2.

[0076] Inductor L1 may provide desired tuning for low-noise operation that simultaneously matches the transistor for low noise and/or gain matching

[0077] Inductor L1 can act as a local “ac ground” that avoids a ground current from flowing through the interconnects. By defining a ground on 148, the interconnect will not impact the differential behavior of the circuitry.

[0078] In other embodiments, inductor L1 may be replaced with a capacitor such as a capacitor C1 the sources of transistor T1 and T2. In other embodiments, inductor L1 may be replaced with an AC ground such as ground GNDr1 or GNDr2.

[0079] In some cases, to accurately apply transistor models, the high-frequency (AC) ground of the chiplets must be established, such as by GNDr1 and/or GNDr2. Heterogeneous integration adds parasitics that change the AC ground. A key feature of the networks 140 is that they can use differential circuitry in the chiplet to establish a local ground. A reactive or resistive source connection network, such as any of the networks 140, creates a well-defined AC ground.

[0080] FIG. 2E is a schematic top view of an electronic assembly 250 that isolates an example of stabilization network circuits 241, 242 and 243 of the interconnection tuning circuits 140 having passive circuitry of chiplets 230, 231 and 232, each having active transistor T1. There may be more or fewer electrical and/or circuit components than shown in for chiplets 230, 231 and 232, wafer 110, and/or stabilization network circuits 241, 242 and 243. The circuitry of chiplets 230, 231 and 232 may be described as having both passive (e.g., R, C, L) and active (e.g., at least transistor T1) devices. Other passive and/or active devices may be in or on the chiplet. Although only three chiplets are shown, there may be numerous chiplets in cavities of wafer 110. There may be between 10 and 1000 such chiplets in the wafer. Each of circuits 241, 242 and 243 may be an isolation stabilization network circuit of chiplets 230, 231 and 232, respectively, in the presence of the interconnect networks extending through interconnects 155 and/or 255.

[0081] Each of chiplets 230, 231 and 232 may represent a version of chiplet 130. Each of circuits 241, 242 and 243 may represent a version of circuitry 140.

[0082] Circuitry 241 may be a network of electronic circuitry that includes at least one resistor R4 between gate G1 and ground GNDr1. Circuitry 241 includes input IN3 electrically attached to a node between gate G1 and resistor

R3; output OUT3 at the drain D1; and a load LD1 at a node between drain D1 and output OUT3.

[0083] Resistor R4 may provide desired DC bias and DC stability to the input IN3 as compared to without the resistor. Resistor R4 may provide desired impedance matching at the input IN3 and output OUT3 of transistor T1 as compared to without the resistor.

[0084] Circuitry 242 may be a network of electronic circuitry that includes at least one resistor R5 in series with one capacitor C8 between gate G1 and output OUT4. Although the resistor is shown closer than the capacitor to the gate, the order of those components may be reversed. Circuitry 242 includes input IN4 electrically attached to a node between gate G1 and resistor R5; output OUT4 at the drain D1; and a load LD1 at a node between drain D1, an end of capacitor C8 and output OUT4.

[0085] Resistor R5 and capacitor C8 may provide stabilization over a range of frequencies between the input IN4 and/or output OUT4 as compared to without the resistor and capacitor. Resistor R5 and capacitor C8 may provide desired DC and/or AC impedance matching at the input IN4 and output OUT4 of transistor T1 as compared to without the resistor and capacitor. Resistor R5 and capacitor C8 may provide series RC feedback between the gate and drain compared to without the series RC feedback in terms of DC and/or AC biasing.

[0086] In some cases, the electrical connection between output OUT4 and the node between capacitor C5 and load LD1 may include passive or active devices.

[0087] Circuitry 243 may be a network of electronic circuitry that includes at least one resistor R6 between output OUT5 and load LD1. Circuitry 243 includes input IN5 electrically attached to gate G1; output OUT5 at a node between the drain D1 and output OUT5; and a load LD1 at the end of the resistor R6 away from the node between drain D1 and output OUT5.

[0088] Resistor R6 may provide desired DC bias and DC stability to the output OUT5 as compared to without the resistor. Resistor R6 may provide desired impedance matching at the input IN5 and output OUT5 of transistor T1 as compared to without the resistor.

[0089] The DC stability and/or impedance matching of stabilization network circuits 241, 242 and 243 produces unconditionally stable chiplets 230, 231 and 232 with respect to signals received on the interconnects 255 from the wafer 110 and thus allows the active device transistor T1 to also be unconditionally stable. It is considered that stabilization of the transistor T1 might incorporate additional lossy components such as resistors and reactive components such as capacitors and/or inductors to stabilize the transistor T1 in a specific frequency band of operation as noted for the transistors. This stabilization can be applied to a single transistor but also to a differential pair of transistors. In this case, both the stability of the differential and common-modes would be guaranteed through the network circuits 241, 242 and 243 implemented on the active chiplets 230, 231 and 232.

[0090] Stabilization network circuits 241, 242 and 243 may also be used on either or both of transistors T1 and/or T2 as shown in FIG. 1C. Any network circuitry of circuitry 140 may be an impedance matching network.

[0091] FIG. 2F shows circuits 260-269 having transistor T1 and/or T2 as RF transistors; with input IN, output OUT, AC ground shown by the triangular ground symbol (which

may also be a DC voltage bias), load LO; switch control signal S; and input output I/O.

[0092] FIG. 2F shows circuits 260 and 261 having transistor T1 as a single-ended RF amplifier with an input at the gate and source, respectively, and an output at the drain.

[0093] FIG. 2F also shows circuit 262 having transistors T1 and T2 in a series connection of devices for an RF cascode with an input at the gate of transistor T1, and an output at the drain of transistor T2.

[0094] Next, FIG. 2F also shows circuit 263 having both of transistors T1 and T2 as a differential RF amplifier with inputs at the gates and outputs at the drains of each transistor.

[0095] Now, FIG. 2F also shows circuit 264 having both of transistors T1 and T2 as a frequency multiplier with inputs at the gates of each transistor and an output tied to the drains of both transistors.

[0096] In addition, FIG. 2F also shows circuit 266 having two sets of transistors T1 and T2 as four total transistors on a single chiplet as an RF FET ring mixer or provider of frequency translation with two inputs IN1 and IN2; two outputs OUT1 and OUT2; and two loads LO1 and LO2.

[0097] Then, FIG. 2F also shows circuit 267 having transistor T1 as an RF switch transistor for an RF switch or mixer with an input at the source, a switch control signal S1 at the gate and an output at the drain.

[0098] Also, FIG. 2F also shows circuit 268 having transistors T1 and T2 in series as RF switch transistors for an RF switch or mixer with an input output I/O at the drain of each transistor and a switch control signal at the gate of each transistor.

[0099] Finally, FIG. 2F also shows circuit 269 having transistors T1 and T2 as part of RF switch transistors for an RF switch or mixer with an input at the source of transistor T1; an output at the drains of transistors T1 and T2; a switch control signal at the gate of each transistor.

[0100] Any of tuning circuitry 140-148 and/or 241-243, such as shown in FIGS. 1C and 2A-2E, may be applied to any of circuits 260-269. Chiplets that exhibit gain as illustrated in 260-263 require stabilization networks to ensure robust operation in the presence of interconnection to the tuning circuits. Chiplets that generate power as illustrated in 260-264 are improved in performance by tuning networks as indicated by the sample in 140-148 and/or 241-243 that produce desirable fundamental or harmonic tuning networks. Switching networks including mixers and RF switches indicate in 265-269 benefit from the network techniques in FIG. 1C to prevent signal power loss on the chiplet and, potentially, due to the interconnection networks.

[0101] FIG. 3A is a schematic cross-sectional view of a device 300 having heterogeneous integration of a radio frequencies (RF) transistor chiplets 130 into a wafer 110, where the chiplets have interconnection tuning circuits 140. Device 300 may include the one or more of the devices of any or all of FIGS. 1A-2F.

[0102] Device 300 may be an electronic assembly having an encapsulation material layer 370 having a top surface 372 and a back surface 374. Device 300 has a host wafer 110 having back surface 112 and front surface 114, with the back surface 112 of the wafer bonded to the top surface 372 of an encapsulation material layer 370 except for cavities 120 in the wafer 110 formed over a plurality of areas 376 of the top surface 372. The cavities may extend from back surface 112, through the wafer and to front surface 114. The cavities have side surfaces 116. The back surface 112 of the wafer may be

directly attached to and touching the top surface 372. The bond between the back surface 112 and the top surface 372 may be a covalent, chemical or atomic bond.

[0103] Chiplets 130 have a backside 134 and a frontside 132, with the backsides 134 of the chiplets 130 bonded directly to at least portion 378 of the plurality of areas 376 of the top surface 372 of the encapsulation material layer. Portion 378 may be the footprint of the chiplet 130 on top surface 372 within the cavity 120. A gap 350 between side surfaces 116 and 136 may be the difference between area 376 and portion 378. The backside 134 may be directly attached to and touching the top surface 372. The bond between the backside 134 and the top surface 372 may be a covalent, chemical or atomic bond.

[0104] The cavities 120 may be through-substrate holes or through substrate holes etched in the wafer at the areas 376. The chiplets 130 may be embedded into the wafer 110 at the substrate holes or at cavities 120.

[0105] A lateral material 360 extends between side surfaces 136 of the chiplets 130 and the side surfaces 116 of the wafer or cavities. The lateral material 360 may mechano-chemically bond the side surfaces 136 of the chiplets 130 to the side surfaces 116 of the wafer. The lateral material 360 may form a mechanical and/or a chemical bond to the side surfaces 136 and to the side surfaces 116. In some cases, the lateral material 360 is a molded material and the bonding is a mechano-chemical bond.

[0106] Material 360 may be a dielectric material. Material 360 may not be a metal and may be an electrical insulator. Material 360 may be or include material that is not conductive, is may not be a semiconductor, may be a plastic, may not be an alloy, may be a bio-material. Material 360 may be an epoxy. It may be an electrical insulator epoxy with electrical insulator particles. It may be epoxy with silica or SiO₂ particles.

[0107] In other cases, the lateral material 360 is not a dielectric material. In these cases, the lateral material 360 may be a metal, a conductor, an alloy or a semiconductor. Material 360 may be an epoxy. It may be a conductive epoxy with conductive particles. It may be epoxy with a metal particles. In these cases, there is a dielectric layer or a space (e.g., of air) between the material 360 and the interconnects 155 and/or 255.

[0108] The lateral material 360 is disposed in gaps 350 between the side surfaces 136 of each of the chiplets 130 and the side surfaces 116 of the corresponding wafer cavity that each chiplet 130 is disposed in. The gap 350 has a width gw of between 1/5 (one fifth) and 10 times a thickness tw of the wafer 110 or chiplets 130.

[0109] The thickness tw of the wafer may be between 20 and 200 microns. It may be between 50 and 125 μ m. It may be 75 μ m. The thickness of one, many or all of the chiplets may be that same as that of the wafer.

[0110] A thickness te of the encapsulation material layer may be between 3 and 100 microns. It may be between 5 and 25 microns. It may be 15 μ m.

[0111] Each of the chiplets 130 have between 3 and 6 sides. They may have 4 sides. The sides may be straight, curved or wavy in profile as viewed from a top perspective. The cavities 120 may have the same number of and sides corresponding to the shapes of the sides of the chiplets 130.

[0112] The encapsulation material layer 370 may be a high-thermal-conductivity backside metallization layer that improves heat transfer from the chiplets 130 to the wafer

110. Layer **370** may be a thermal plane that improves heat conduction away from the chiplets by increasing thermal conduction from the chiplets **130** and to layer **370** and/or wafer **110**. Layer **370** be a material in direct contact with the chiplets **130** to increase thermal conduction between the materials of the chiplets **130** and that of layer **370**. In some cases, the encapsulation material layer **370** has a coefficient of thermal expansion between those of or equal to one of those of the wafer **110** and of the chiplets **130**.

[0113] In other cases, there is no encapsulation material layer **370** and the chiplets **130** are bonded to the wafer **110** using the lateral material. the chiplets **130** may be bonded to the wafer **110** using only the lateral material.

[0114] Interconnects **155** may be formed directly on or may be formed over (e.g., formed on a dielectric or air gap over) the lateral material **360** and connect electrical (e.g., power, ground and/or signal) contacts **138** of the chiplets **130** to contacts **118** of the wafer **110**. Interconnects **155** may include direct interconnect routing or traces that is formed directly on the lateral material (e.g., without any dielectric/air gap), and that extends from the chiplet contacts **138** to wafer contacts **118** and electrical routing. The interconnect routing **155** may include low loss high-performance DC, RF, and mm-wave routing from the chiplet contacts **138**, directly on the lateral material **360**, and to wafer contacts **118**. Interconnects **155** may be directly on material **360** by being bonded to and/or directly attached to (e.g., touching) the top surface of the lateral material **360**.

[0115] In some cases, wafer **110** includes an electronic integrated circuit (not shown), at least one integrated circuit contact **118** (e.g., contact pad) formed on the front wafer surface **114**, and at least one through-wafer cavity **120** having side surface **116** that join back surface **112** to front surface **114**. In some cases, a chiplet **130** is held in the through-wafer cavity **120** by a lateral material **360** that attaches at least one side surface **116** of the through-wafer cavity **120** to at least one side surface **136** of the chiplet **130**. In some cases, lateral material **360** fills gap **350** of the cavity, thus attaching most of the side surfaces **136** of chiplet **130** to the side surfaces **116** of through-wafer cavity **120**; however material **360** does not attach the backside **134** of chiplet **130** to top surface **372** of layer **370**.

[0116] A passivation layer (not shown) can be arranged on most of the front surface **114** of wafer **110**. Conducting vias (e.g., TWVs) arranged through the passivation layer can connect the active and/or passive circuitry of wafer **110** to contacts **118** (e.g., contact pads) on front surface **114**. Wafer **110** can be a silicon wafer or substrate, which allows taking advantage of known fabrication processes and manufacturability on large wafer diameters.

[0117] It is noted that wafer **110** can include any integrated circuit, active or passive, made possible by a chosen manufacturing process; for example, a CMOS manufacturing process. In some cases, the thickness of the one or more integrated circuit layers can for example be only a fraction of the thickness tw of wafer **110** (for example between $\frac{1}{10}$ and $\frac{1}{1000}$ of the thickness of wafer **110**; for example 50 nm thick with a wafer 50 μ m thick). In some cases, the thickness of wafer **110** can be reduced after fabrication of integrated circuits of the wafer and for example before etching the through-wafer cavity **120** or after filling gap **350** with lateral material **360**.

[0118] Chiplet **130** may include one or more transistors having its terminals connected to at least one integrated

circuit contact **138** (e.g., contact pad), such as by a conductive via (not shown). Chiplet **130** can comprise a substrate and integrated circuit layers formed on top of its substrate, the thickness of the integrated circuit layers being for example only a fraction of the thickness of the substrate (for example between $\frac{1}{10}$ and $\frac{1}{1000}$ of the thickness of the substrate). In some cases, the total thickness of chiplet **130** is smaller than the total thickness of host wafer **110**. In some cases, lateral material **360** contacts the side surfaces **136** of chiplet **130** along most of their height (at least 50% of the height, starting from close to the top surface of chiplet **130**). Preferably, lateral material **360** contacts essentially all of the side surfaces **136** of chiplet **130**. Preferably, lateral material **360** fills completely gap **350**, up to a level essentially flush with the front surface **114** of host wafer **110**.

[0119] In some cases, lateral material **360** and/or layer **370** holds the chiplet **130** such that the chiplet frontside **132** is flush with the front surface **114**. Being “flush” may be understood as meaning that the two surfaces are in a same plane, or have, with respect to each other, a small or negligible height difference. The two surfaces may be flush, such as resulting from the process of permanently attaching chiplet **130** to the side surfaces **116** of through wafer cavity **120** while both the chiplet frontside **132** and the front surface **114** are attached temporarily to an adhesive laminate **340**. The two surfaces may be flush, such as resulting from polishing or CMP of those surfaces after removing the temporarily to an adhesive laminate **340**.

[0120] It is considered that the host wafer **110** can be vertically diced at dicing lines (shown by the vertical bars in FIGS. 3A and 3B) along a perimeter **386** of the wafer around at least one chiplet to form a chip having the at least one chiplet and an area of the wafer surrounding the at least one chiplet.

[0121] FIGS. 3A and 3B may show how the interconnect structures **155** and **255** may sit directly on top of and touch the lateral material **360** or other dielectric material between the chiplets **130** and wafer **110**. The electrical interconnects electrically connect the passive devices of the wafer and the transistors of the chiplets. The interconnect between the chiplet and wafer may mistune the transistor, such as by introducing reactive mismatches in stability, AC signal, DC signal, impedance, frequency and/or other electronic signal matching with the input and/or output to the chiplet or a transistor on the chiplet. To avoid this mistuning, networks on the chiplet produced a robust characteristic. A benefit of any one or more of the interconnection tuning circuits **140** is that it reduces this electrical mistuning of passive devices and the transistors caused by the electrical interconnects that may impact circuit performance and reduces the uncertainty of the electrical interconnects to improve the circuit performance.

[0122] Also, chiplets **130** are preferably pre-tested to verify their functionality and/or performance. As a result, the yield of the final device **300** or diced devices is much improved over integration of component chips, in which the functionality of the component chips is not verified until after integration.

[0123] Advantageously, by allowing different electrical component chiplets **130** to be manufactured separately from each other and from wafer **110**, electronic components of those in chiplets **130** and wafer **110** can be tested separately before assembling them. In case one of the components of a certain electrical component chiplet **130** or of wafer **110**

has poor fabrication yields, it is possible to replace the component to improve the fabrication yield of the assembled electrical component to produce a completed product device 300 or die thereof, having the chiplets 130 together in the cavities of the wafer 110. For example, If an electrical component of a certain chiplet type of the chiplets 130 has poor fabrication yield, it can be rejected and replaced with a different chiplet to improve the fabrication yield of that electrical component chiplet type without spending time and money to remake the entire assembly of wafer 110, to produce a completed product device 300 or die thereof, having the chiplets 130 and wafer 110.

[0124] Further, because embodiments allow fabricating different electrical component ones of chiplets 130 separately from each other and from wafer 110, all of the component types of chiplets 130 and wafer 110 do not need to be exposed to steps in the fabrication of all the different electrical component ones of chiplets 130 that could potentially damage other ones of chiplets 130 or damage wafer 110.

[0125] Thus, embodiments can reduce manufacturing costs by using small component chips in chiplets 130 having specific features and made of exotic expensive materials, in combination with integrated circuits of other chiplets 130 and/or wafer 110 having more common features and made of cheaper common materials.

[0126] According to embodiments, chiplets 130 can include a GaN, InP or GaAs electrical component and can be fabricated on a substrate such as Si, SiGe, InP, GaAs, Alumina, or diamond. In some cases, electrical components or integrated circuits of host wafer 110 can comprise metal routing and passive components fabricated at the wafer scale. In some cases, interconnection 155 can be made using conductors made out of thin films, thick, plated interconnects, multi-layers, etc. The interconnections can for example be made using the back-end steps of a manufacturing process.

[0127] FIG. 3B is a schematic cross-sectional view of a device 301 having heterogeneous integration of a radio frequencies (RF) transistor chiplets 130 onto a wafer 310, where the chiplets have interconnection tuning circuits 140. Device 301 may include the one or more of the devices of any or all of FIGS. 1A-2F. As compared to wafer 110, wafer 310 does not have cavities 120, but instead has the chiplets mounted on a top surface of wafer 310.

[0128] In this embodiment, the chiplets 130 are mounted on or bonded to the top surface 114 of the wafer 310 instead of being in cavities 120. In this case, there may be no cavities for the chiplets. The interconnects 155 and 255 between the chiplets and wafer are on or above side surface 136 instead of on or above lateral material 360. However, the interconnects may have the same mistuning noted. Here, the concepts above also apply regarding existence of interconnects 155 and 255 between the chiplets and wafer, mistuning of those interconnects, the various interconnect tuning circuits 140 for those interconnects, and the issues and advantages provided by the interconnect tuning circuits, such as noted herein.

Closing Comments

[0129] Throughout this description, the embodiments and examples shown should be considered as exemplars, rather than limitations on the apparatus and procedures disclosed or claimed. Although many of the examples presented herein

involve specific combinations of method acts or system elements, it should be understood that those acts and those elements may be combined in other ways to accomplish the same objectives. With regard to flowcharts, additional and fewer steps may be taken, and the steps as shown may be combined or further refined to achieve the methods described herein. Acts, elements and features discussed only in connection with one embodiment are not intended to be excluded from a similar role in other embodiments.

[0130] As used herein, “plurality” means two or more. As used herein, a “set” of items may include one or more of such items. As used herein, whether in the written description or the claims, the terms “comprising”, “including”, “carrying”, “having”, “containing”, “involving”, and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of”, respectively, are closed or semi-closed transitional phrases with respect to claims. Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. As used herein, “and/or” means that the listed items are alternatives, but the alternatives also include any combination of the listed items.

It is claimed:

1. An electronic assembly comprising:

a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet;

the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and

electrical interconnects between the at least one chiplet and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein:

the stabilization network includes: a first resistor between a first input of a first transistor device and a biasing node, a second resistor between a second input of a second transistor device and the biasing node, wherein the node is to a first AC ground and DC voltage that provides a first DC bias to the first and second inputs.

2. The electronic assembly of claim 1, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.

3. The electronic assembly of claim 1, wherein the interconnects electrically connect the passive devices and the transistors,

wherein the at least one chiplet has passive tuning circuits to reduce electrical mistuning of the transistors caused by the electrical interconnects that degrades transistor performance and increase uncertainty of the circuit operation, and

wherein each chiplet is within a through wafer cavity of the wafer.

4. The electronic assembly of claim 1, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is 180 degrees out of phase with the first signal phase.

5. An electronic assembly comprising:

a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet;

the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and

electrical interconnects between the chiplets and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein:

the unilateralization or gain boosting network includes: at least one capacitor between a first input of the first transistor and a second output of the second transistor.

6. The electronic assembly of claim 5, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.

7. The electronic assembly of claim 5, wherein the interconnects electrically connect the passive devices and the transistors, and

wherein the passive tuning circuits reduce electrical mistuning of the transistors caused by the electrical interconnects that degrades transistor performance and increase uncertainty of the circuit operation.

8. The electronic assembly of claim 5, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is 180 degrees out of phase with the first signal phase.

9. The electronic assembly of claim 5, wherein each chiplet is within a through wafer cavity of the wafer.

10. An electronic assembly comprising:

a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet;

the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and

electrical interconnects between the chiplets and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein: the power delivery network includes: at least one capacitor between a gate of the first transistor and a first AC ground and DC voltage that provides a first DC bias to the gate of the first transistor.

11. The electronic assembly of claim 10, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.

12. The electronic assembly of claim 10, wherein the interconnects electrically connect the passive devices and the transistors, and

wherein the passive tuning circuits reduce electrical mistuning of the transistors caused by the electrical interconnects that degrades transistor performance and increase uncertainty of the circuit operation.

13. The electronic assembly of claim 10, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is 180 degrees out of phase with the first signal phase.

14. The electronic assembly of claim 10, wherein each chiplet is within a through wafer cavity of the wafer.

15. An electronic assembly for heterogeneous integration of radio frequencies (RF) transistor chiplets having interconnections to or between tuning circuits, the assembly comprising:

a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet;

the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and

electrical interconnects between the chiplets and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein:

the low-noise network includes at least one inductor between a source of the first transistor and a source of the second transistor.

16. The electronic assembly of claim 15, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.

17. The electronic assembly of claim 15, wherein the interconnects electrically connect the passive devices and the transistors, and

wherein the passive tuning circuits reduce electrical mistuning of the transistors caused by the electrical inter-

connects that degrades transistor performance and increase uncertainty of the circuit operation.

18. The electronic assembly of claim **15**, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is 180 degrees out of phase with the first signal phase.

19. The electronic assembly of claim **15**, wherein the wafer includes at least one of resistors, capacitors, inductors, through substrate vias, dielectric layers, metal layers; and wherein the wafer includes at least one layer of silicon (Si), silicon dioxide (SiO₂), silicon carbide (SiC), quartz, or Gallium Arsenide; and wherein each chiplet includes at least two RF transistors and the interconnects from the transistors

to contact pads on a frontside surface of the chiplet; and wherein each chiplet is a high-end pre-fabricated transistor chiplet.

20. The electronic assembly of claim **15**, wherein the wafer has front and back surfaces, through wafer cavities in the wafer formed through a plurality of areas of the top surface of the wafer, the through wafer cavities having side surfaces of the wafer; wherein the at least one chiplet is a plurality of chiplets each having a backside and a frontside disposed in the through wafer cavities, each of the chiplets having the second circuit;

further comprising a lateral material between side surfaces of the chiplets and the cavity side surfaces of the wafer, wherein the lateral material bonds the side surfaces of the chiplets to the cavity side surfaces of the wafer; and

wherein the electrical interconnects are over the lateral material and between the chiplets and wafer.

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